Stereoisomerism

Stereoisomers :- Isomers which have same structural formula but have diff arrangement of atoms or groups in space (3D).

Classification of stereoisomerism :-

i) Configurational → Enantiomerism, Diastereomers (Separable, Non interconvertible)

ii) Conformational (interconvertible, Non-separable)

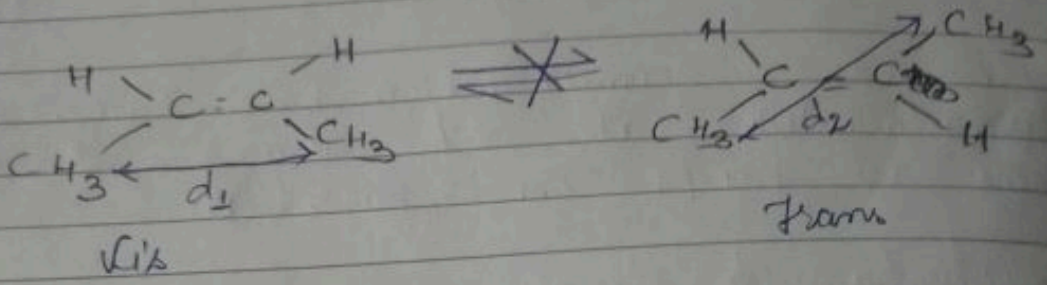
Configurational → Geometrical iso. / Optical iso.

→ Configurational isomers : Stereoisomers which are not interconvertible into each other. (They are convertible by breaking and reformation of bonds).

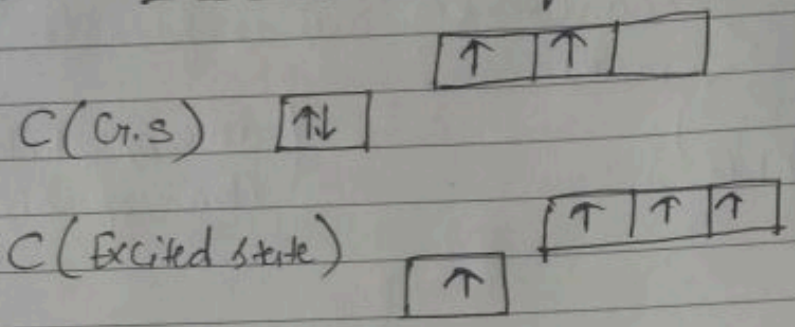
Geometrical isomerism :- Stereoisomers which are non-interconvertible into each other due to presence of restricted rotation.

They have different distance b/w two functional groups.

2-Butene $CH_3 - \overset{sp^2}{CH} = \overset{sp^2}{CH} - CH_3$ (Structural formula)



3D Structure of C=C





Conditions of Geometrical isomerism

1. Geometry of compound should be contain min 2 types of relative distance b/w two position of same atom

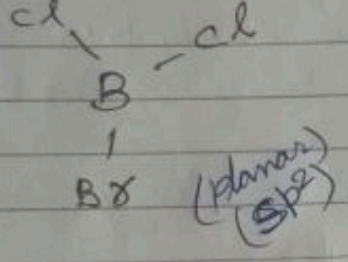
2. Compounds should be contain double bond (C=C , C=N , N=N) or

Substituted ring (restricted rotation).

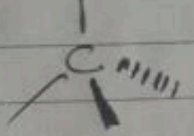
3- Both corners of double bond should be contain 2-2 different groups.

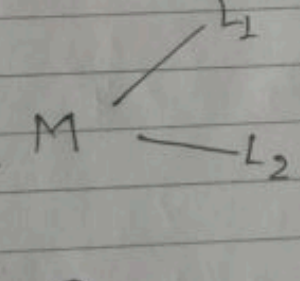
eg.1 $x - C = C - y$ (G.I = not possible)

eg.2 $\rangle C = C \langle$, $x\ y$ $C = C$ $x\ y$, $x \neq y$, G.I = ✓

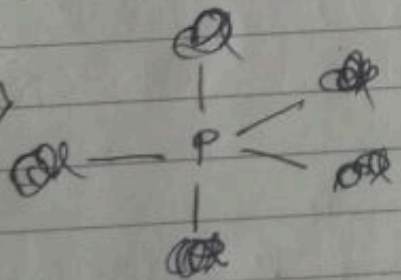
eg.3) 

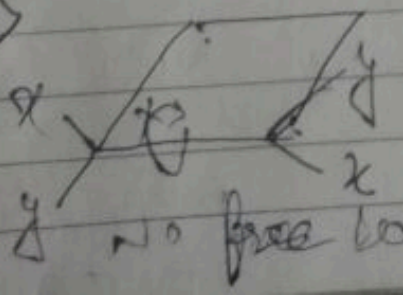
G.I (Not possible)

eg.4 
G.I (Not possible)

eg.5 

Square planner
(G.I = possible)

eg.6) 
This geometry can show geometrical isomer.

eg.7) 

G.I possible
No free rotation

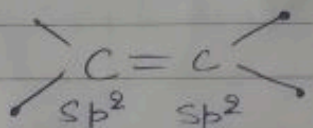
Types of geometrical centres :-

[A] Geo. iso in C=C containing compounds.
[B] " " " C=N " " "
[C] " " " N=N " " "
[D] " " in substituted rings.
[E] " " " in cyclo alkene due to C=C of ring.

(F) " " " in cumulene.
g) " " " in Spiro comp.
h) " " " in Bicyclo comp. etc

A] Geo. iso in C=C containing compounds

⇒ structure



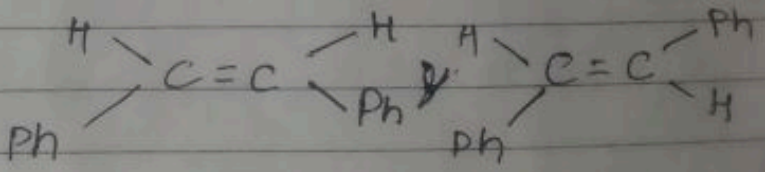
$d_1 \neq d_2$

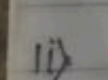
Condition for G.I $xy\,C=C\,xy$, $x \neq y$ G.I = ✓

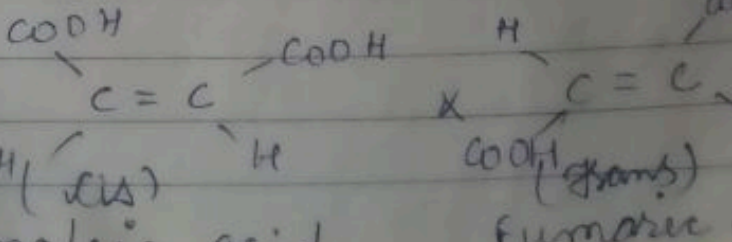
$xy\,C=C\,ab$, $x \neq y$, $a \neq b$ } G.I ✓

as:

i) Ph - CH = CH - Ph

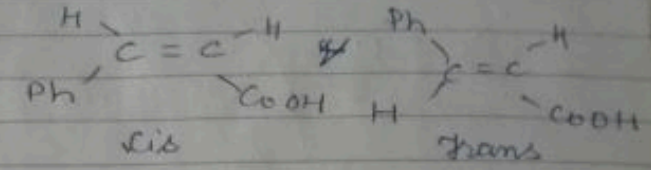


ii)  - HC = CH - COOH

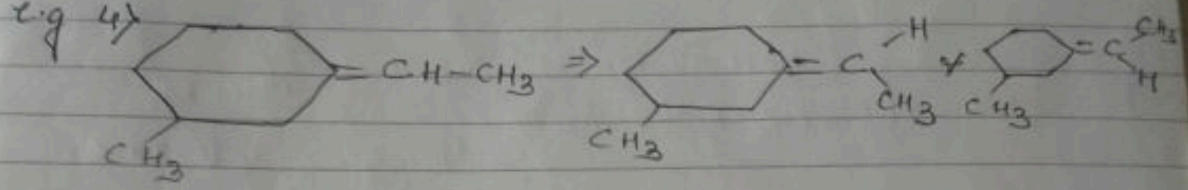


OH Butendioic acid

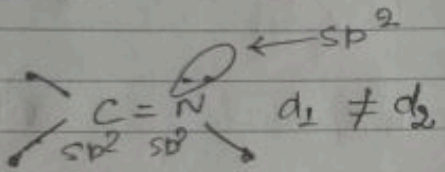
$Ph-CH=CH-COOH$
Cinnamic acid



e.g 4)

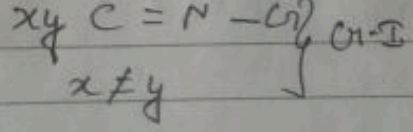


B) Cred. sig in C=N
Structure

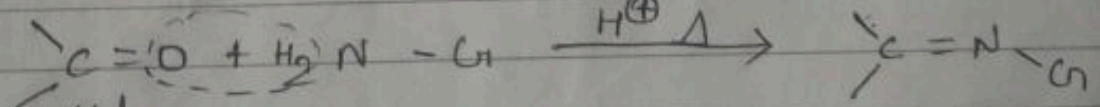


$d_1 \neq d_2$

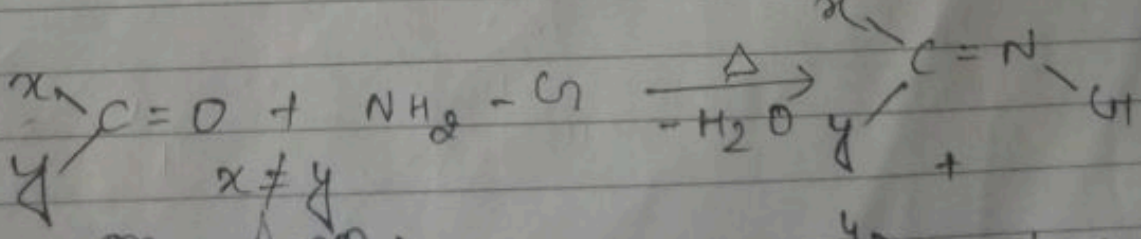
Condition for G.I



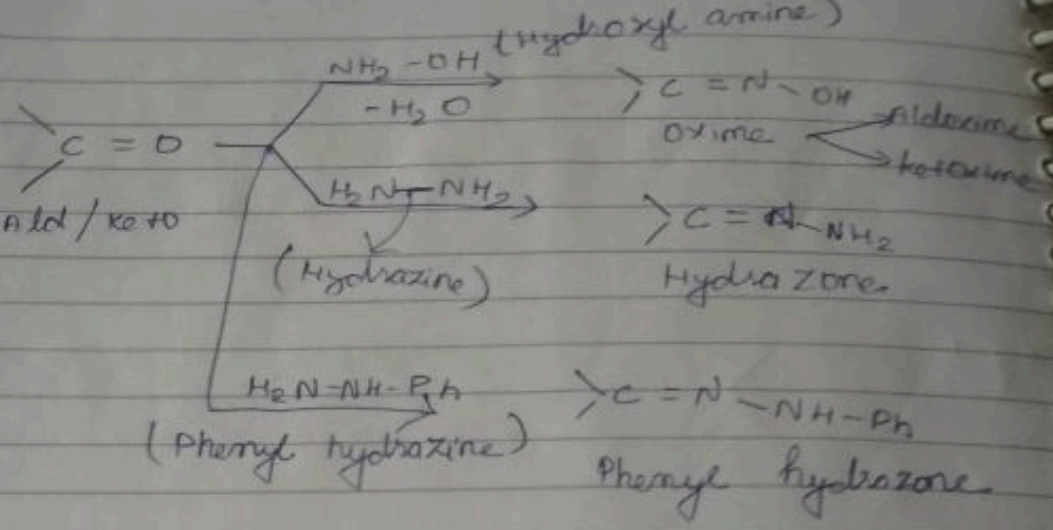
formation of C=N :-



Note:

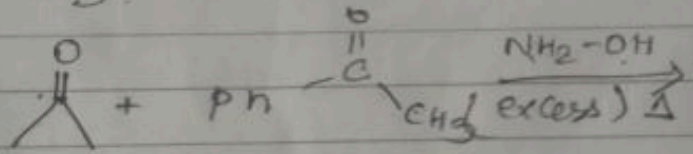


on hydrolysis.
carbonyl comp.

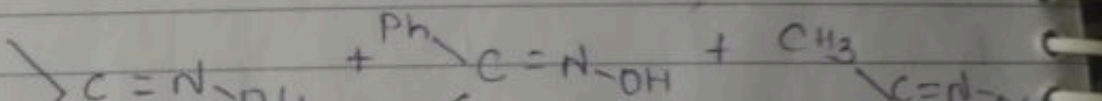


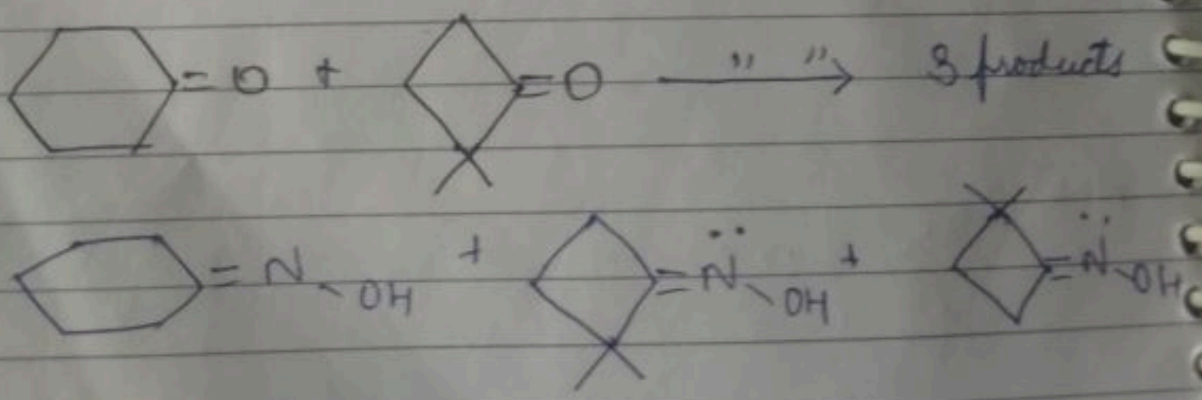
Q. Identify no. of oxime, products for following reactions.

e.g :- $CH_3-CHO \xrightarrow[\Delta]{H_2N-OH}$ 2 products $\left[\begin{matrix} CH_3 \\ H \end{matrix} \!\!>\!\! C=N-OH \right]$

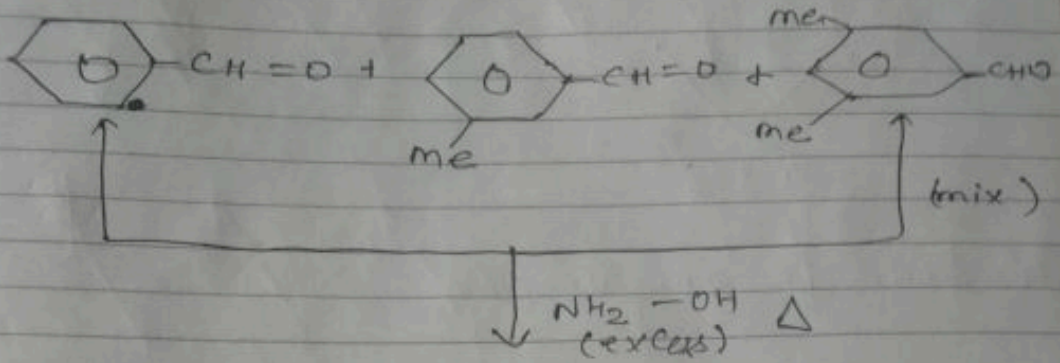
eg 2) 

1 + 2 = 3 products

eg 3) 

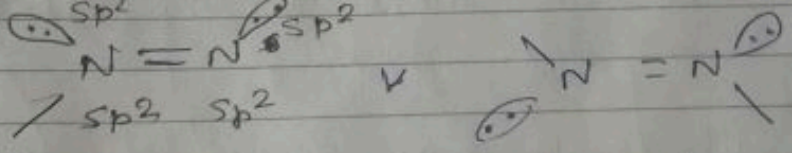
eg 3) 

eg 4>



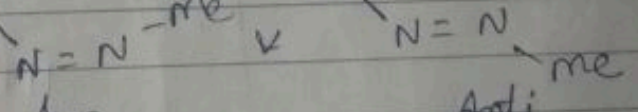
2 + 2 + 2 = 6 products

C> Che. Test in $-N=N-$ containing comp.



Syn form Anti form

eg.1> $me-N=N-me$ ⇒ Azomethane

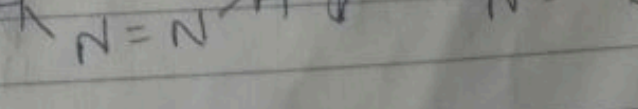


Syn Anti

e> $Ph-N=N-Ph$ ⇒ Azobenzene

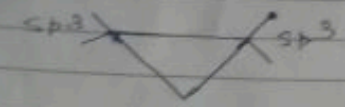


3> $H-N=N-H$ ⇒ Diazene



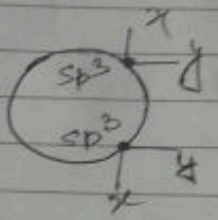
D) C.I in substituted rings:-

Structure



$d_1 \neq d_2$

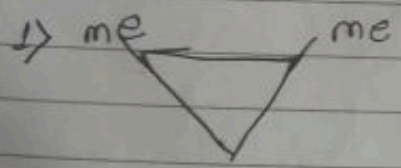
Condition for geo. isomerism



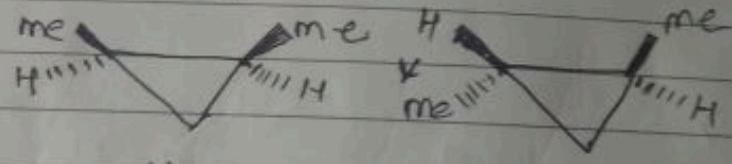
$x \neq y$ G.I = 2

Minimum two sp^3 atoms of ring should contain 2-2 diff. groups.

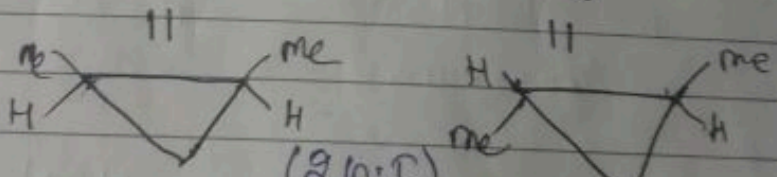
Q. Draw all possible geo. isomers of given comp.

1) 

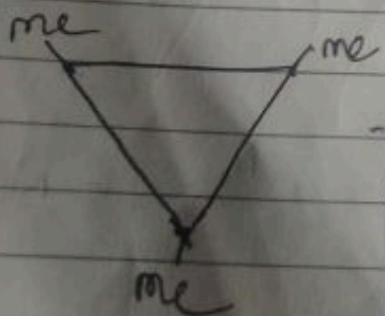
Geo. iso 2

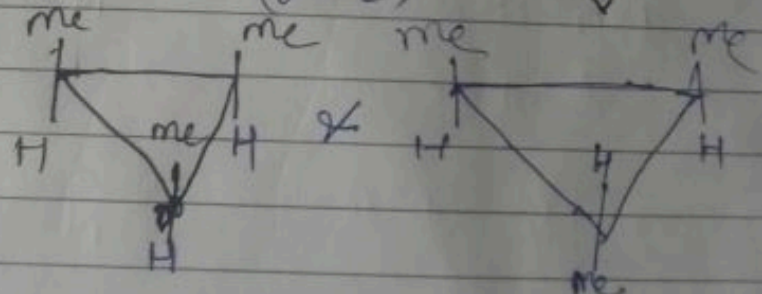


II



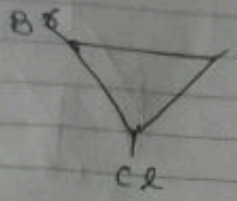
(2 G.I)

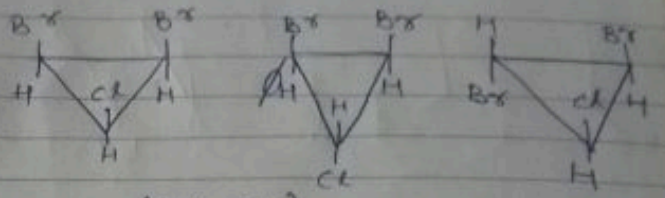
2. 



(2 G.I)

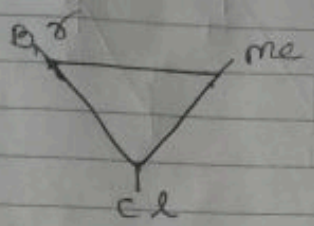
3)

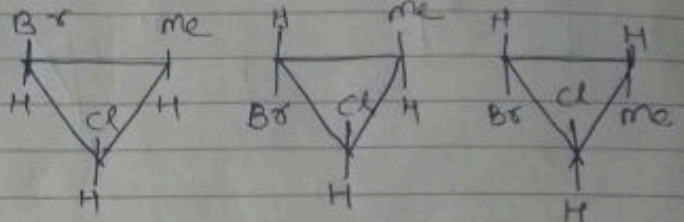


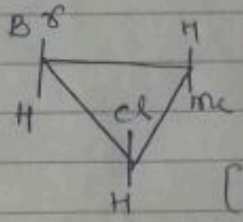


(3 G.I)

4)

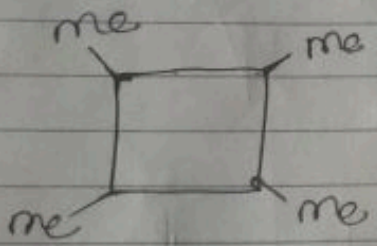


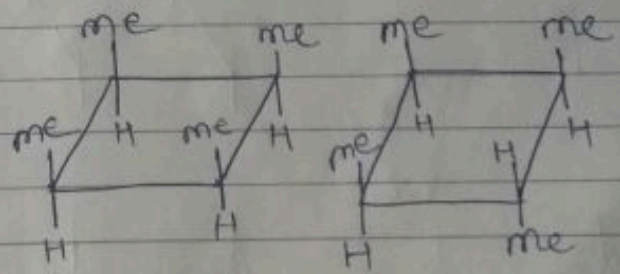


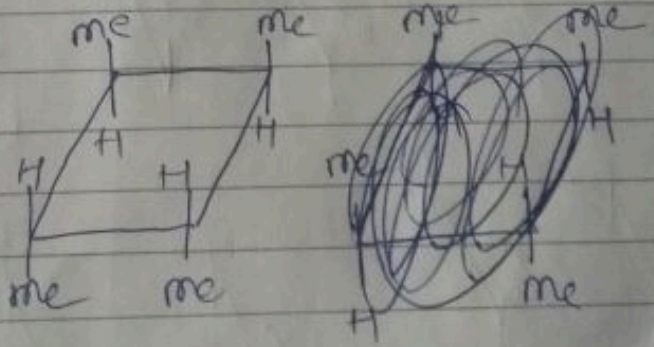


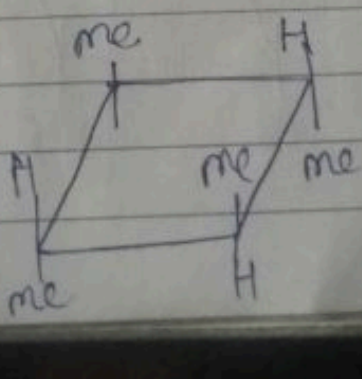
[G.I = 4]

5)

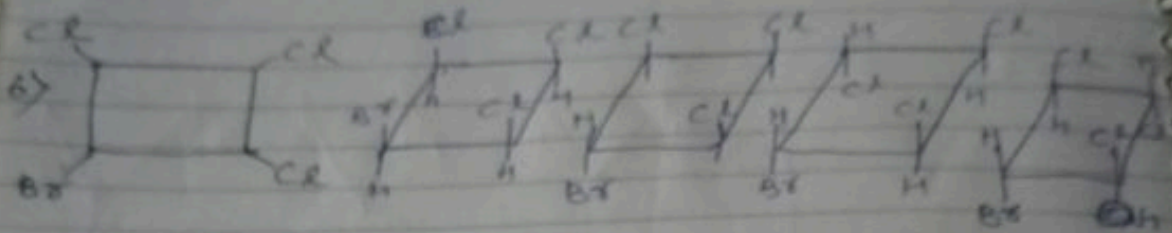


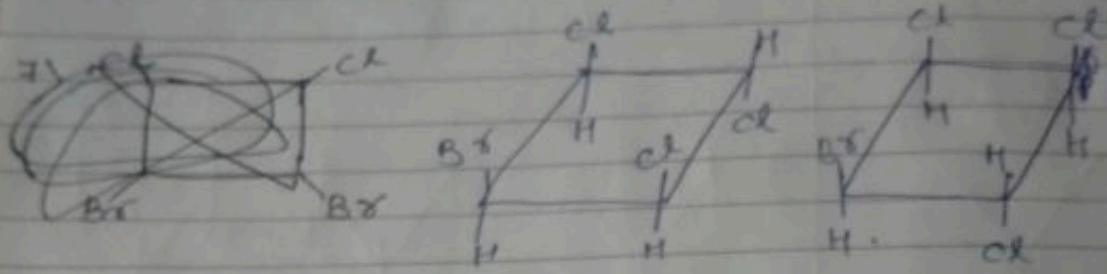


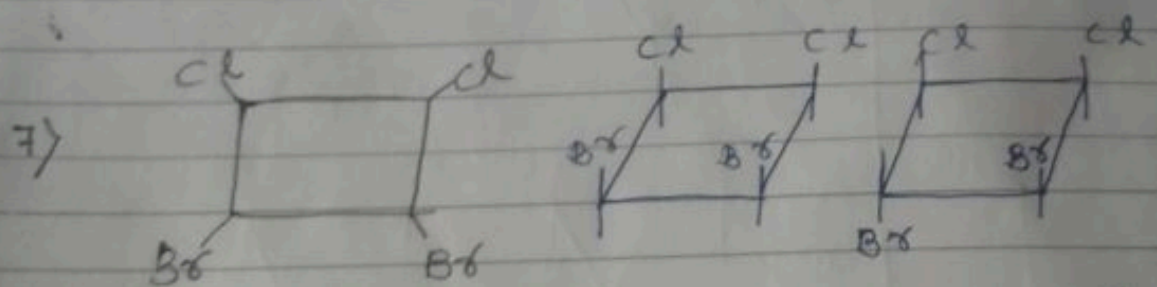


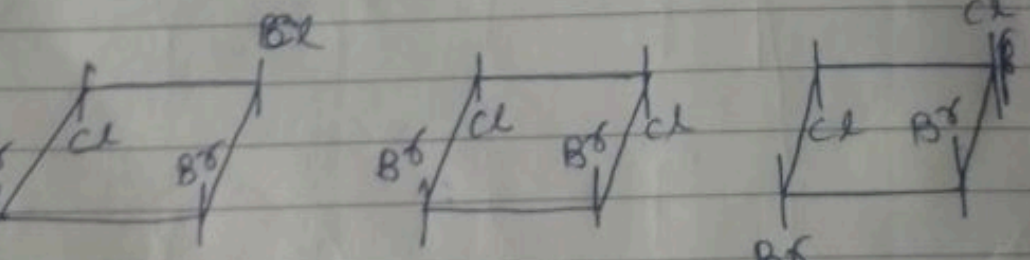


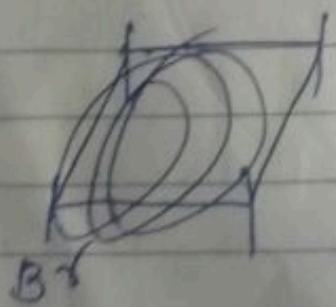
[4 G.I]

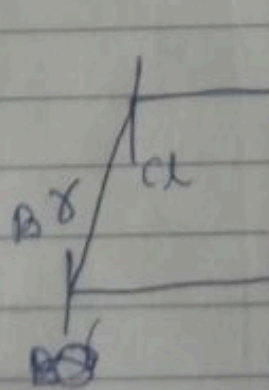




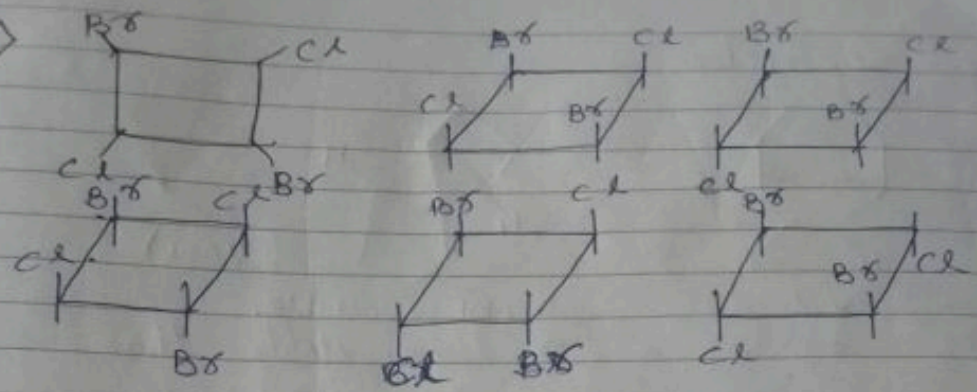








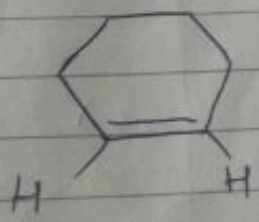
(6)

8) 

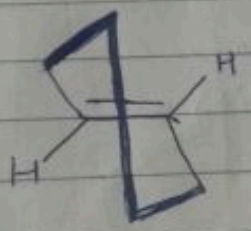
6) Geometrical isomerism in cycloalkenes (due to C=C of ring):

On small cycloalkenes (up to 7 membered) geo. isomerism not possible because in small cycloalkenes only cis form exist in nature but its trans form are unstable due to high strain.

Eg) Cyclohexene

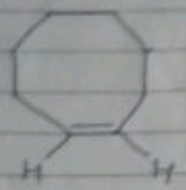


cis

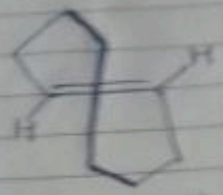


(unstable due to high ring strain)

eg 2) cyclooctene

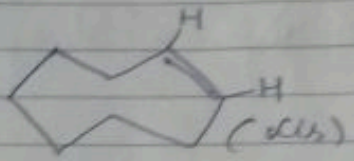


cis

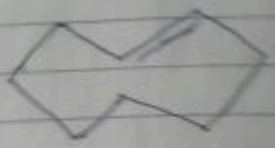


(trans)

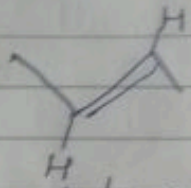
⇒ not isolatable
⇒ lesser stable than cis isomer.



(cis)



(trans)



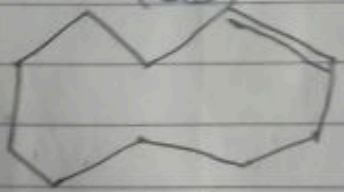
(trans)



(cis)

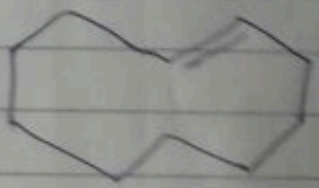
eg 3) cyclodecene

(cis)

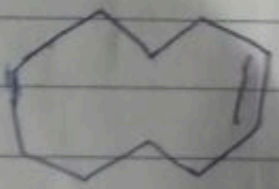


(cis)

(trans)



II

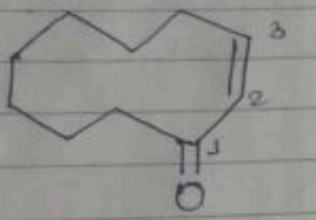


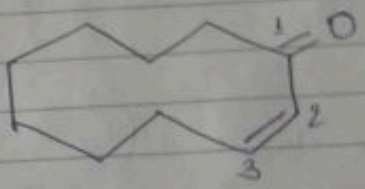
Stability

Ring size	In form	Transform
$C_3 - C_7$	exist in nature	NOT exist
$C_8 - C_{11}$	more stable	less stable
C_{12} - higher	lesser stable	more stable

Q. what is relation b/w them

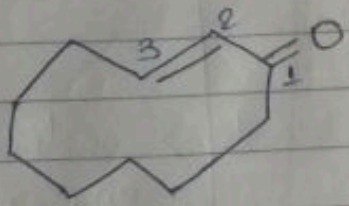
a)

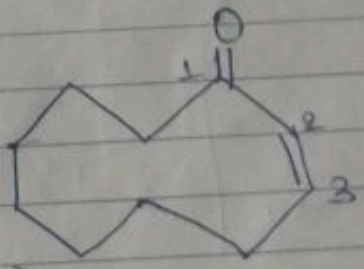




(identical)

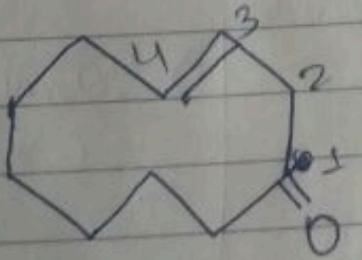
b)

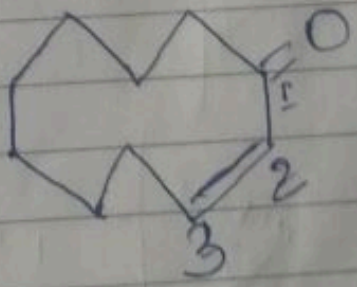




(Geo. isomers)

c)

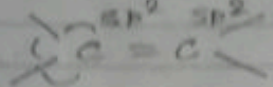




(Positional iso)

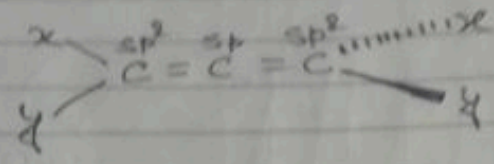
1) Geo isomerism is due to C=C in cumulene

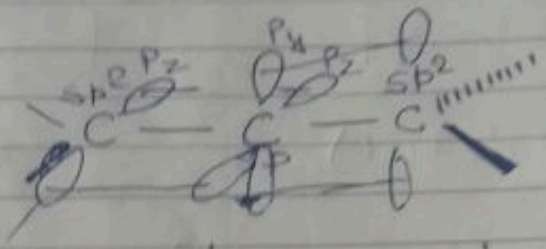
Continuous C=C
(Cumulene containing polymers)



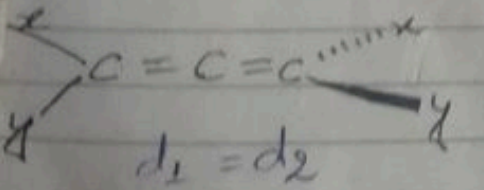
⇒ Structure of cumulene

e.g.:-

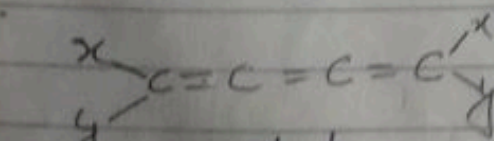




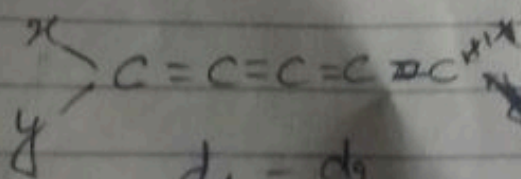
⇒ Structure



$d_1 = d_2$



$d_1 \neq d_2$



$d_1 = d_2$

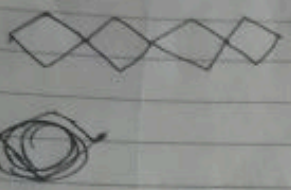
Terminal (x, y)	Geo. iso
1°	Not possible
in same plane	possible
1°	NOT possible

⇒ Conditions for geo. iso in cumulene

i) Terminal ($x \neq y$)

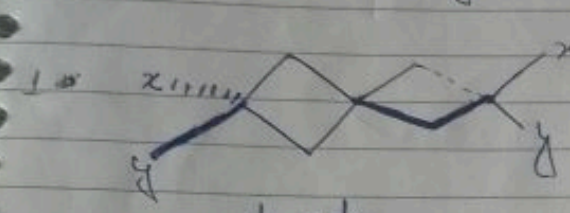
ii) odd no of cumulative

(vi) Geometrical isomerism in spiro comp.:



Two adjacent spiro rings are 1° to each other.

1.

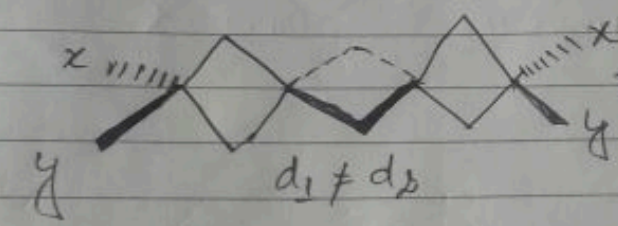


Terminal
$x \neq y$
$\Rightarrow \perp$

Cn. I

Not possible

2.



$x \neq y$

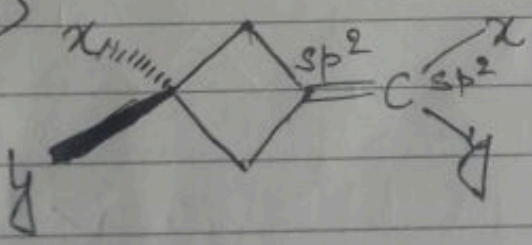
or
Same
Plane

Conditions

i) Terminal ($x \neq y$)

ii) odd no. of spiro rings

eg 3)

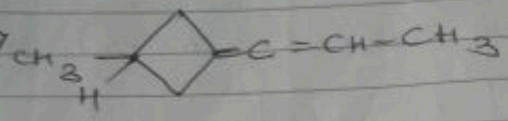


Cn. I = X

eg 4)

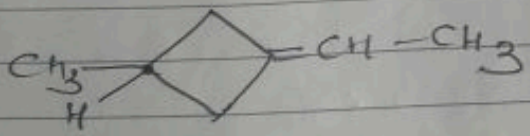


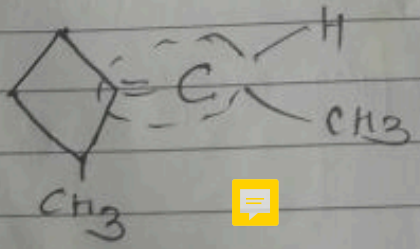
(odd) Cn.S ✓

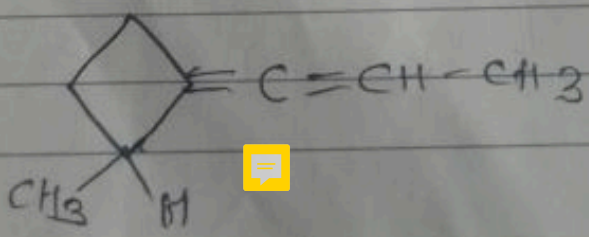
eg 5)  Cn.S ✓

eg 6)  Cn.S ✓

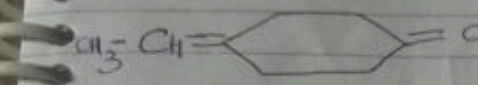
eg 7)  Cn.S ✗

eg 8)  Cn.S ✗

eg 9)  Cn.S = ✓

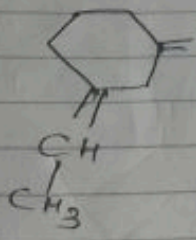
eg 10)  Cn.S ✓

eg 11)



(G.I ✗)

eg.12)

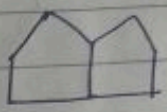


(G.I ✓)

11) Geo. iso in bicyclo compounds

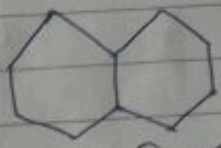
on Bicyclic compounds geo. isomerism possible when both rings have min 5-5 atoms.

eg.1)



G.I = ✓

eg 2)



G.I ✓

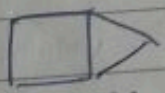
decalin

eg 3)



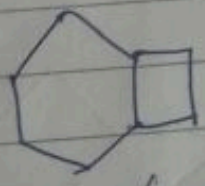
G.I = ✓

eg 4)



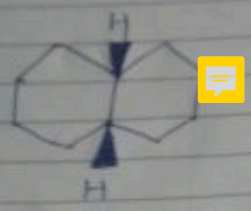
G.I ✗

eg.5)

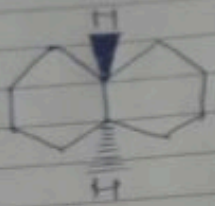


G.I = ✓ (exceptions) अगर 6 या उससे ज्यादा ring हो और दूसरी ओर की हो तो भी G.I show होता है।

In decalene

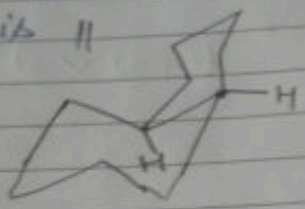


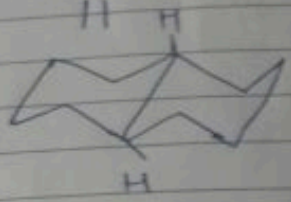
Cis



Trans

Cis





trans

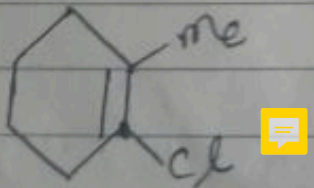
Q. Identify compounds which can show geo. isomerism.

1)  NO G.I

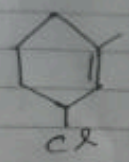
2) Me − C ≡ C − Et NO G.I

3)  G.I ✓

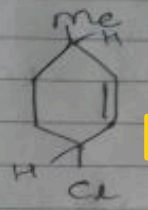
4) Ph − N = N − CH_3 G.I ✓

5)  G.I ✗

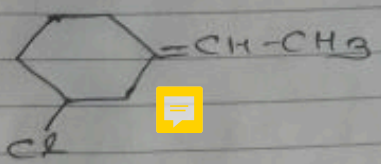
double bond along G.I

6)  see cond^n

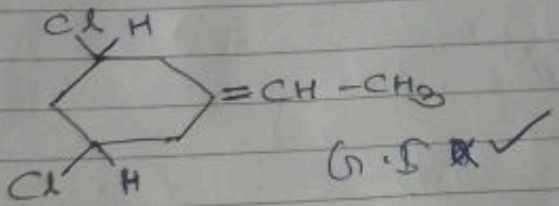
G.I ✗

7) 

G.I ✓

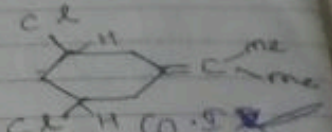
8) 

G.I ✓

9) 

G.I ✗ ✓

↳ एक double bond के along G.I नहीं होगा क्योंकि Ring के about G.I होगा

10) 

G.I ✗

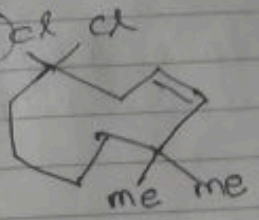
Same like 9

11) 

G.I ✓

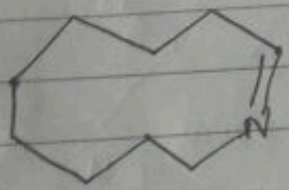
12) 

G.I ✓

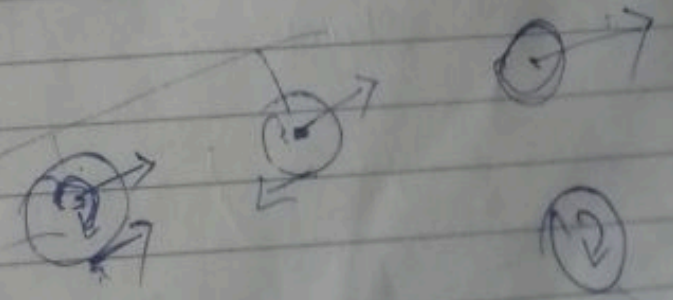
13) 

G.I ✗ ✓

↳ along double bond

14) 

G.I ✓



Configurational nomenclature of Stereochemical centres

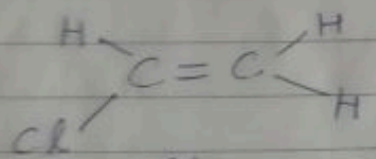
A) Relative configuration
- i) → Cis-trans nomenclature
- ii) → Syn-Anti " "

B) Absolute configu
- → E, Z Nomenclat

A i) Cis trans Nomenclature

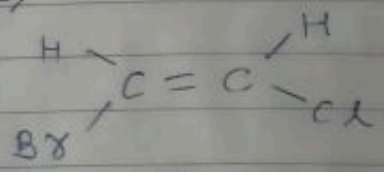
Cis = same group same side
Trans = same group opposite side

eg :1



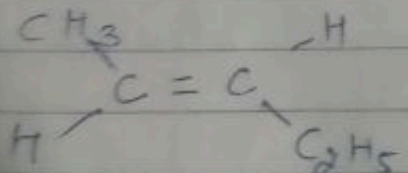
Cis

eg :2)



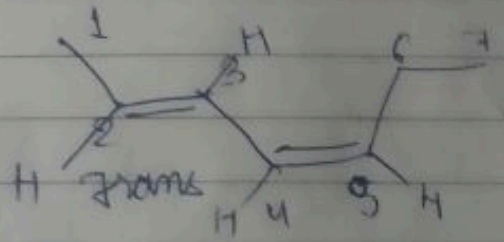
Cis

eg 3)



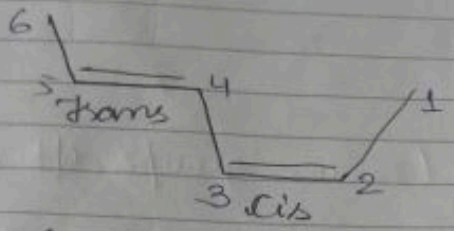
Trans

eg 4)



(2-Trans - 4-(Cis) Hepta-2,4-dien

eg =>



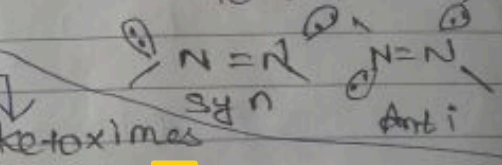
Cis > trans

(2-Cis, 4-trans) Hexa-2,4-diene

Syn-Anti Nomenclature

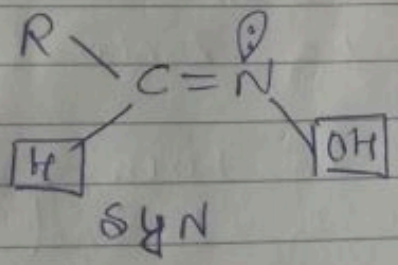
in oximes

in $-N=N-$ (with respect to lop-lop)

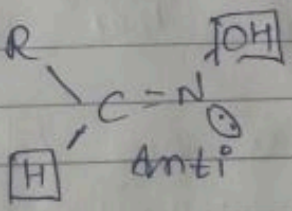


Aldoximes

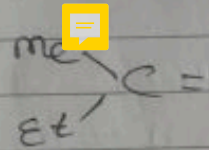
Ketoximes



Syn



Anti



Syn-methyl ketoxime

or

Anti methyl ketoximes

Aldoximes में syn Anti H और OH से define होता है।

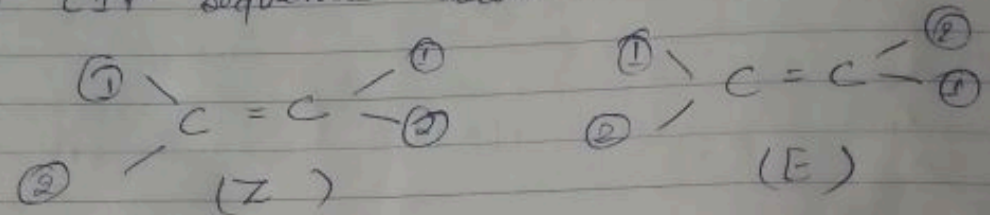
Absolute configuration

E-Z Nomenclature

E = Entagen = opposite ⇒ (Senior groups off side)

Z = Zussamene = together ⇒ (" " same side)

Give priority order (as 1, 2) of two groups of both corners of geometrical centres according to CIP sequence rule.



CIP (Chan - Ingold - Prelog) Rule

1. The group in which the first atom as higher atomic no that will be senior

2. where first atoms are same in both groups then considers atomic no. of second atom (so on).

3. where different isotopes of same element present on same position in diff group then heavier isotope containing group will be senior.

4) Bond pair > lone pair

5. When multiple bonds are in a group then draw a hypothetical equivalent structure according to following way.

eg.1 $-CH=CH_2 \Rightarrow -CH-CH_2$ (C) (C)

eg.2) $-\underset{\underset{H}{|}}{C}=O \Rightarrow -CH(-O-C=O)$

eg.3) $-C\equiv C-H \Rightarrow -C(=O)-C(-C)=C$

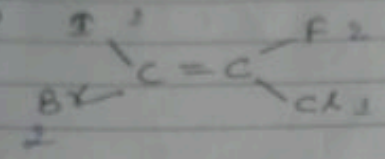
eg.4) $-C\equiv N \Rightarrow -C(=N-C)-N-C$

eg.5) $-CH=N-CH_3 \Rightarrow -CH(-N-C)-N-CH_3$

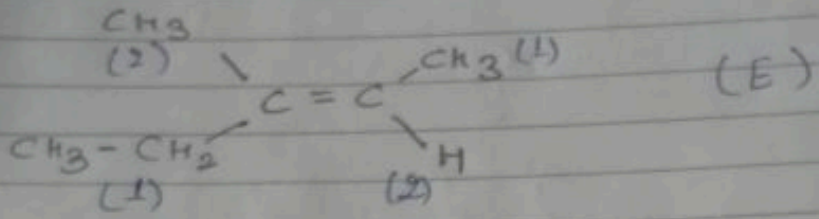
eg.6) $-N=C=O$ (isocynate) $\Rightarrow -N(-C)-C(=O)-O-C$

eg.7) $-CH=CH-\overset{\overset{O}{\|}}{C}-CH_3 \Rightarrow -CH(-C)-CH(-C)-C(=O-C)-CH_3$

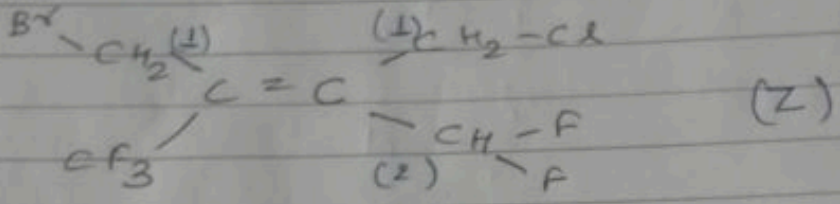
Q. Identify absolute configuration (E, Z) for following

1) 

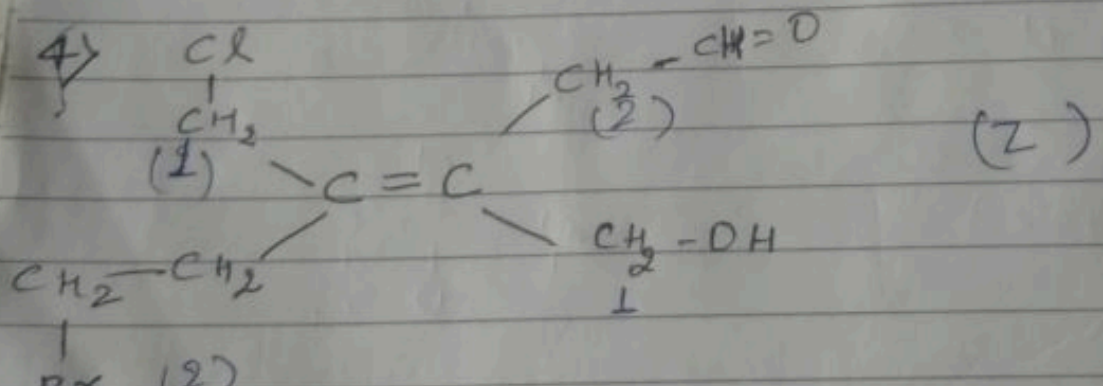
(E)

2) 

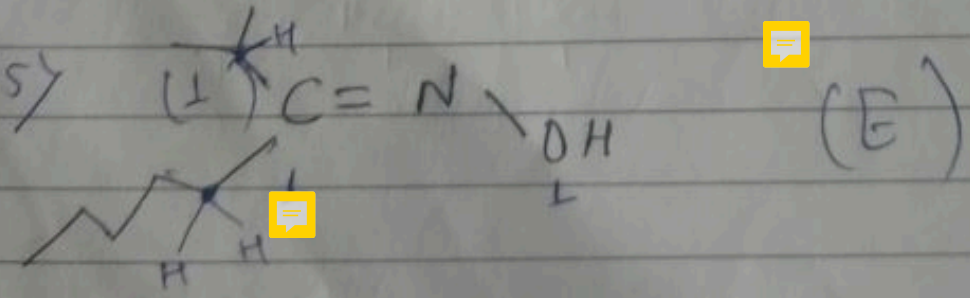
(E)

3) 

(Z)

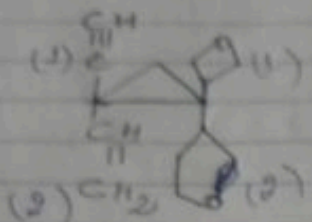
4) 

(Z)

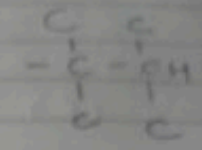
5) 

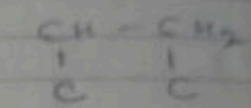
(E)

6)



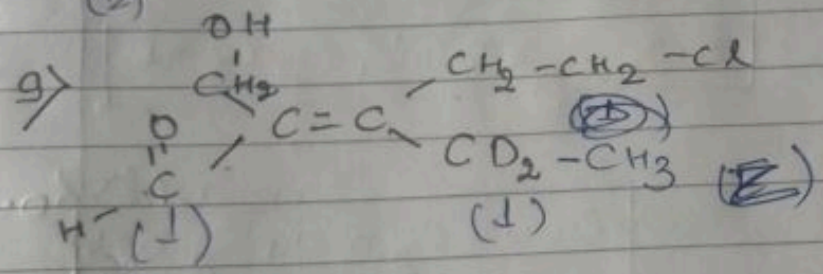
(Z)

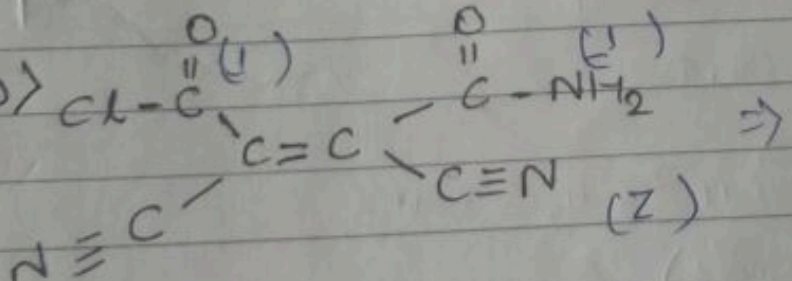
$-C\equiv CH$ = 

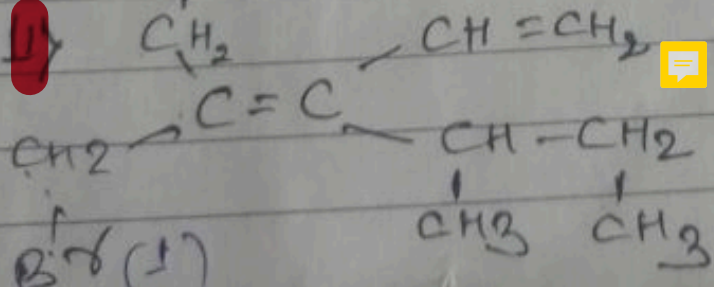
$CH=CH_2$ = 

7) $CH_3-C(CD_3)=C(D)-CH_2-Cl$ (E)

8) $CH_3-CH_2-C(D)=C(CH_2-CD_3)-CD_2-CH_3$ (E)

9) 

10) 

11) 

In IUPAC Nomenclature and Priority Rules:
. cis > trans
. Z > E

1>



(Z > E)

(2.Z, 4E) Hexa 2 4 diene

Calculation of no. of geometrical isomers.

For dissimilar ends containing comp.

no. of geo. iso = 2^n

n = no. of centres which can show geo. isomers

For similar ends containing comp.

when n = even no.

$G.I = 2^{n-1} + 2^{n/2 - 1}$

n = odd no.

$G.I = 2^{n-1} + 2^{\frac{n-1}{2}}$

Q. Identify no. of G.I

$CH_3 - CH = CH - CH = CH - CH_2 - CH_3$

(Dissimilar ends)

$n = 2$

$\Rightarrow 2^n = 2^2 = 4$

2> $CH_3-CH=CH-CH=CH-CH_3$ (similar ends)

$\Rightarrow n = 2$

$\Rightarrow G.I = 2^{2-1} + 2^{\frac{2}{2}-1}$

$= 2^1 + 2^0 = 3$

1st dob — 2nd dob

Cis — Cis

Trans — Trans

Cis — Trans

3. $CH_3-CH=CH-CH=CH-CH=CH-CH_3$

(Similar)

$n = 3$

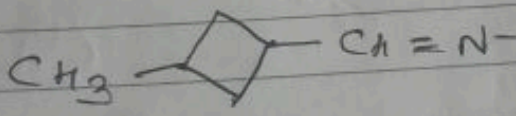
$G.I = 2^{n-1} + 2^{\frac{n-1}{2}}$

$= 2^2 + 2^1 = 6$

4> $CH_3-CH=CH-CH=CH-CH=CH-Cl$

$2^n = 2^3 = 8$ (dissimilar ends)

5>



$CH=N-OH$ $n=2$ dissimilar ends

$2^2 = 4$

6> $CH_3-CH=N$



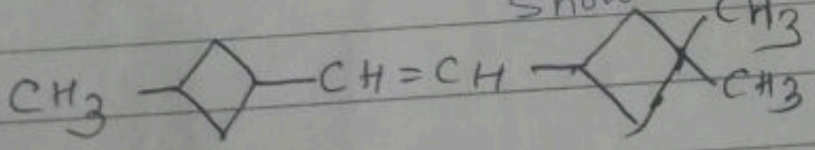
$N=CH-CH_3$ (Similar ends)

$n = 3$ $2^{n-1} + 2^{\frac{n-1}{2}}$

Show ring to about 60.2 pm

$= 2^2 + 2^1 = 5$

7>



(dissimilar ends)

$n = 2$ $2^2 = 4$

Here, cumulene is odd, so plane is same.

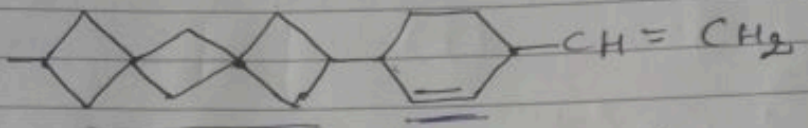
2) $CH_3-CH=C=C=C=CH$ 

$n = 3$

dissimilar ends

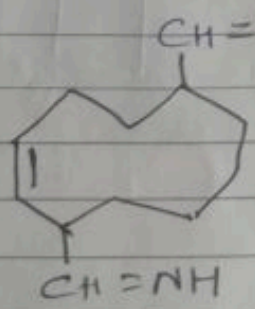
$2^3 = 8$

Q. identify no. of geometrical isomers.

1) 

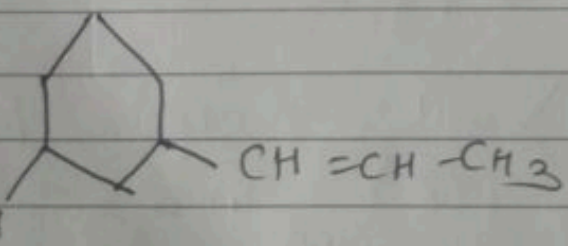
$n = 2$ (dissimilar ends)

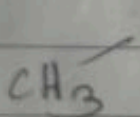
$2^2 = 4$

2) 

$n =$ (dissimilar ends)

$2^4 = 16$

Q 3) 

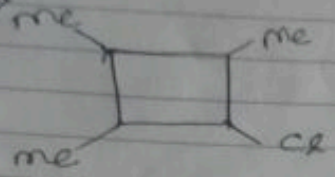


similar ends

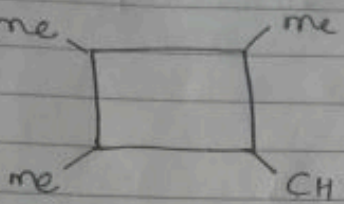
$n = 3$

$2^{n-1} + 2^{\frac{n-1}{2}} = 2^2 + 2^1 = 6$

Q. Identify no. of geometrical isomers.

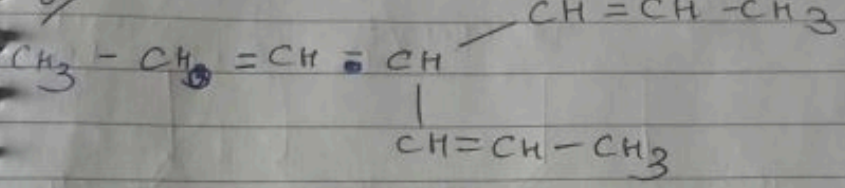
4) 

G.I = 6

5) 

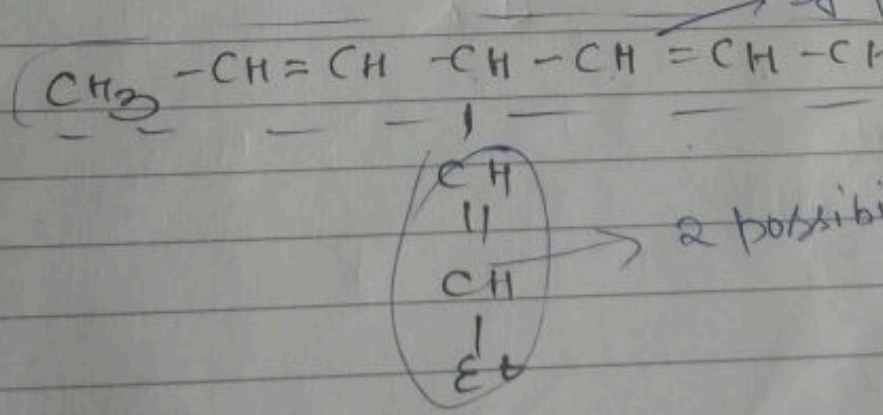
G.I = 6 × 2 = 12

6)



1) Cis - Cis - Cis
2) Trans - Trans - Trans
3) Cis - Trans - Trans
4) Trans - Cis - Cis

G.I = 6

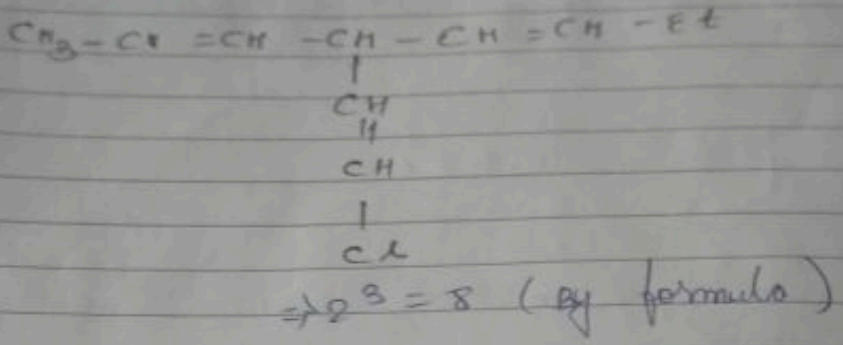
7) 

By formula = 3

2 possibilities

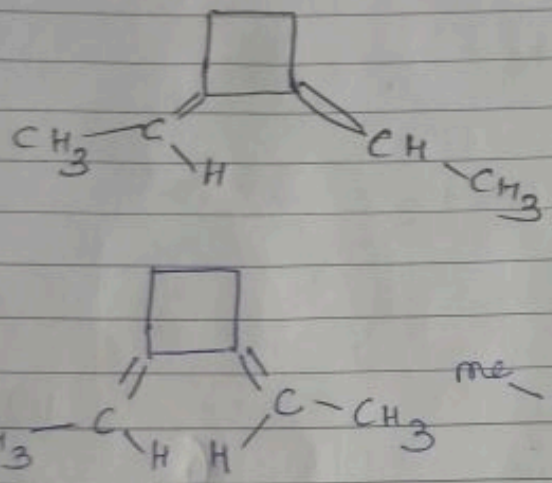
Total = 3 × 2 = 6

8)



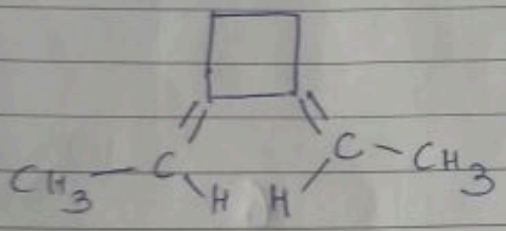
$=> 2^3 = 8$ (By formula)

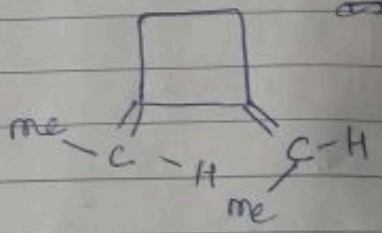
9)

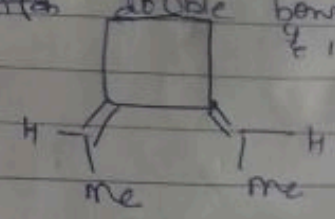


$6 \cdot 5 = 3$

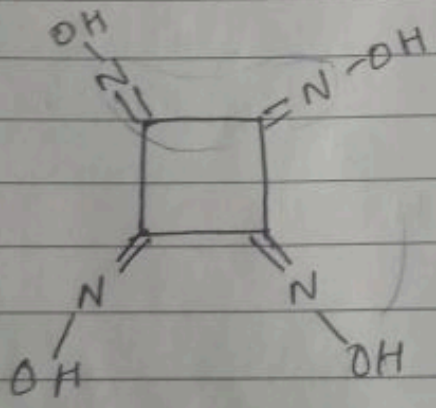
यहाँ पर दोनों उपर या [illegible] उपर [illegible] का [illegible] नहीं [illegible] क्योंकि double bond है।







10)



Ans. 4

* Physical properties of Geometrical Isomers

1. Boiling point of geo. Iso $\propto$ Dipole moment
2. Melting point of G.I $\propto$ Symmetrical structure

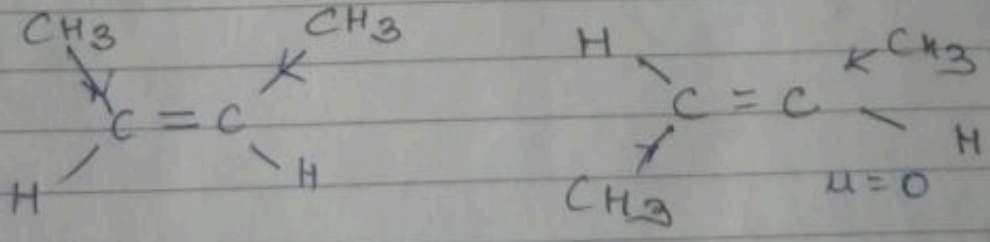
⇒ More Symmetrical ~~molecules~~ isomers have strong packing in solid state so they have ∞ greater melting point.

⇒ Generally Trans isomers have greater melting point.

3. Solubility in water (of G.I) $\propto$ Dipole moment
4. Stability of G.I $\propto \frac{1}{\text{Vander wall strain}}$

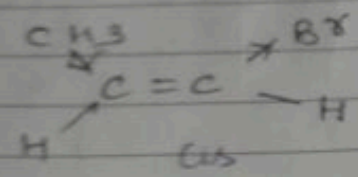
Repulsion b/w bulky group

eg.1)

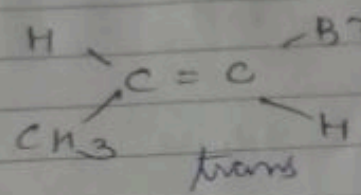


1.	Dipole moment	Cis > Trans
2.	Boiling point	Cis > Trans
3.	Melting point	Trans > Cis
4.	Solubility in water	Cis > Trans
5.	Stability	Trans > Cis

eg 2)



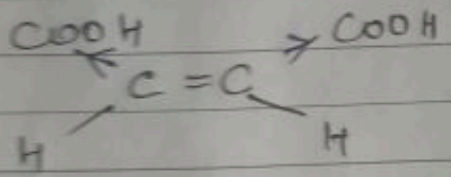
cis



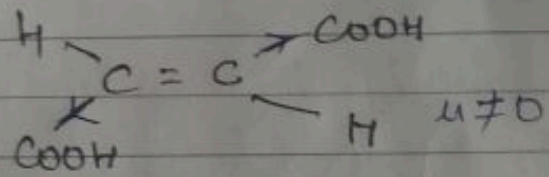
trans

1. Dipole moment	cis < trans
2. Boiling point	cis < trans
3. Melting point	trans > cis
4. Solubility in water	cis < trans
5. Stability	trans > cis

eg 3)



Maleic acid



Fumaric acid

1. Dipole Moment	cis	> trans
2. Boiling point	cis	< trans (By H-bond)
3. Melting point	cis	< trans
4. Solubility in water	cis	> trans ([redacted])
5. Stability.	cis	< trans